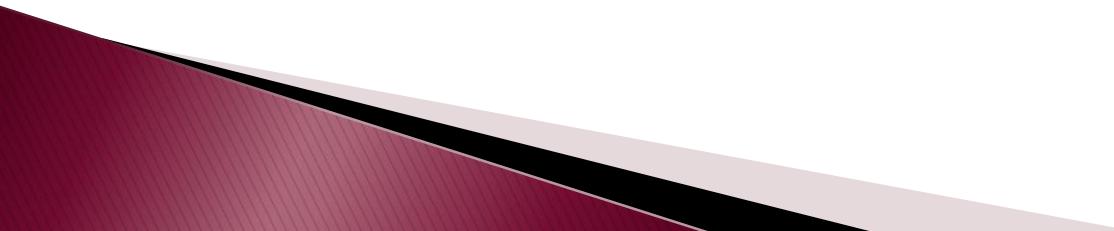


Introduction to programming

Lab02

Revision

- ▶ Variables
 - ▶ Types
 - ▶ Type conversion
 - ▶ Operators
 - ▶ Input
 - ▶ Output
 - ▶ Math module
- 

Homework 1

- ▶ Installation
- ▶ Math module

Homework 2

- ▶ Input the radius of a circle. The program has to calculate its circumference and area. (use the `math.pi` constant defined in the `math` module).

Solution

```
import math

r = int(input("Radius:"))
a = math.pi * r ** 2
c = 2 * math.pi * r
print("Area:", a)
print("Circumference:", c)
```

Homework 3

- ▶ Input two integers. Write a program that calculate the arithmetic and geometric mean of the numbers.

Solution

```
import math
```

```
a = int(input("a = "))
```

```
b = int(input("b = "))
```

```
print("Arithmetic mean: ", (a+b)/2)
```

```
print("Geometric mean: ", math.sqrt(a*b))
```

Format function

- > `print("Pi value = {:.2f}".format(math.pi))`
- > `print("Pi value = {0:.2f}".format(math.pi))`
- > `print("Pi value = {pi:.2f}".format(pi = math.pi))`
- > `print("Name = {}, price = {}".format("apple", 990))`
- > `print("Name = {0}, price = {1}".format("apple", 990))`
- > `print("Name = {name}, price = {price}".format(name = "apple", price = 990))`
- > `print("{0:d} = {0:b} = {0:o} = {0:X}".format(15))`

Number systems

- ▶ Decimal number system
 - the concept of place value
- ▶ Numbers can generally be described in the following form:

$$a_n a_{n-1} \dots a_1 a_0 a_{-1} a_{-2} \dots a_{-m}$$
$$\sum_{i=-m}^n a_i \cdot 10^i, \text{ when } 0 \leq a_i < 10.$$

- ▶ In a base- p ($p > 1$) number system, the digits used are 0, 1, ..., $p-1$, and the place values are powers of p :

$$\sum_{i=-m}^n a_i \cdot p^i, \text{ when } 0 \leq a_i < p.$$

Example: 123,45

Number systems

- ▶ Binary (base two) number system
 - digits: 0, 1
- ▶ Ternary (base three) number system
 - digits: 0, 1, 2
- ...
- ▶ Octonary (base eight) number system
 - digits: 0, 1, 2,..., 7
- ...
- ▶ Decimal (base ten) number system/ten base
 - digits: 0, 1, 2,..., 9
- ...
- ▶ Hexadecimal (base sixteen) number system
 - digits: 0, 1, 2, ..., 9, A, B, C, D, E, F

Binary p = 2	Ternary p = 3	Quinary p = 5	Octonary p = 8	Decimal p = 10	Duodecimal p = 12	Hexadecimal p = 16
0	0	0	0	0	0	0
1	1	1	1	1	1	1
10	2	2	2	2	2	2
11	10	3	3	3	3	3
100	11	4	4	4	4	4
101	12	10	5	5	5	5
110	20	11	6	6	6	6
111	21	12	7	7	7	7
1000	22	13	10	8	8	8
1001	100	14	11	9	9	9
1010	101	20	12	10	a	A
1011	102	21	13	11	b	B
1100	110	22	14	12	10	C
1101	111	23	15	13	11	D
1110	112	24	16	14	12	E
1111	120	25	17	15	13	F
10000	121	26	20	16	14	10

Number systems base names

- 2 binary
- 3 ternary
- 4 quaternary
- 5 quinary
- 6 senary
- 7 septenary
- 8 octonary
- 9 nonary
- 10 decimal
- 11 undenary
- 12 duodecimal
- 13 tridecimal
- 14 quattuordecimal
- 15 quindecimal
- 16 hexadecimal

Convert to base 10 (p→10)

Generally:

$$\begin{aligned} & a_n a_{n-1} \dots a_0 . a_{-1} a_{-2} \dots a_{-m} \text{ }_p = \\ & a_n \cdot p^n + a_{n-1} \cdot p^{n-1} + \dots + a_0 \cdot p^0 + a_{-1} \cdot p^{-1} + \\ & \quad + a_{-2} \cdot p^{-2} + \dots + a_{-m} \cdot p^{-m} \\ & 0 \leq a_i < p \end{aligned}$$

For example:

$$\begin{aligned} 263.15_7 &= 2 \cdot 7^2 + 6 \cdot 7^1 + 3 \cdot 7^0 + 1 \cdot 7^{-1} + 5 \cdot 7^{-2} = \\ &= 98 + 42 + 3 + \frac{1}{7} + \frac{5}{7^2} = 143 \frac{12}{49}_{10} \end{aligned}$$

Exercise – Homework

- Convert each of the following representations from other bases into their equivalent **base ten** form:

1. $1011100.101_2 =$

2. $1221.12_3 =$

3. $152.43_6 =$

4. $173.104_8 =$

5. $841.47_9 =$

6. $13A.2F_{16} =$

Convert base-10 numbers to other bases ($10 \rightarrow p$)

$113.45_{(10)}$

113	2
56	1
28	0
14	0
7	0
3	1
1	1
0	1



0	.45	2
0	.90	
1	.8	
1	.6	
1	.2	
0	.4	
0	.8	
1	.6	
1	.2	

$1100001.01110011_{(2)}$

Exercise – Homework

- ▶ Convert each of the following base-ten representations into **binary** form:
 1. $262_{10} =$
 2. $32.3_{10} =$
 3. $171.64_{10} =$
- ▶ Convert each of the following base-ten representations into **octonary** form:
 1. $51_{10} =$
 2. $71.8_{10} =$
 3. $117.18_{10} =$
- ▶ Convert each of the following base-ten representations into **hexadecimal** form:
 1. $33_{10} =$
 2. $81.9_{10} =$
 3. $218.12_{10} =$

Connection between the octal and the binary system

Binary	Octal
000	0
001	1
010	2
011	3
100	4
101	5
110	6
111	7

Connection between the hexadecimal and the binary system

Binary	Hexadecimal	Binary	Hexadecimal
0000	0	1000	8
0001	1	1001	9
0010	2	1010	A
0011	3	1011	B
0100	4	1100	C
0101	5	1101	D
0110	6	1110	E
0111	7	1111	F

Exercise – Homework

- ▶ Convert each of the following binary representation to their equivalent octal and hexadecimal forms.
- ▶ Convert each of the following hexadecimal representations to their equivalent binary and octal forms:

1. $1110\ 1001\ 1100\ 0011_2 =$

2. $1011\ 0111\ 0101\ 0100_2 =$

3. $1000\ 1101\ 1111\ 0011\ 1101_2 =$

4. $1010\ 1011\ 0011\ 1110\ 0001\ 0101_2 =$

5. $3BCF_{16} =$

6. $BF29_{16} =$

7. $48C5_{16} =$

8. $63AE_{16} =$

Webpages – Practicing – Converters

- ▶ <http://planetcalc.com/862/>
- ▶ <https://www.rapidtables.com/convert/number/base-converter.html>

Format function

```
> print("Binary {0:b} decimal = {0:d}" .format(0b101))
> print("Octal {0:o} decimal = {0:d}" .format(0o17))
> print("Hexadecimal {0:x} decimal = {0:d}" .format(0xab))
> print("Hexadecimal {0:X} decimal = {0:d}" .format(0XB2F))

> print("{0} binary = {0:b}".format(5))
> print("{0} binary = {0:b}".format(123))
> print("{0} octal = {0:o}".format(12))
> print("{0} hexadecimal = {0:x}".format(12))
> print("{0} hexadecimal = {0:X}".format(12))
```

Functions

- ▶ **bin()** – is an in-built function that takes an integer x and returns the binary representation of x as a string.
 - `>>> bin(15)`
 - `'0b1111'`
- ▶ **oct()** – is an in-built function that takes an integer x and returns the octonary representation of x as a string.
 - `>>> oct(15)`
 - `'0o17'`
- ▶ **hex()** – is an in-built function that takes an integer x and returns the hexadecimal representation of x as a string.
 - `>>> hex(15)`
 - `'0xf'`
- ▶ **int()** function converts between number systems. The first parameter is a string, the second is the base (2..36).
 - `>>> int('101',2)`
 - `5`
 - `>>> int('17',8)`
 - `15`
 - `>>> int('A9C4',16)`
 - `43460`

Bitwise operators

Operator	Description
&	Bitwise and
	Bitwise or
^	Bitwise xor
~	Bitwise not
<<	Bitwise left shift
>>	Bitwise right shift

Logical operators

Operator	Description
and	Logical and
or	Logical or
not	Logical negation

Comparison operators (relational operators)

Operator	Description
==	Equal to
!=	Not equal to
>	Greater than
<	Less than
>=	Greater than or equal to
<=	Less than or equal to

Compound assignment operators

Operator	Example	Equal to
=	a = 10	a = 10
+=	a += 10	a = a + 10
-=	a -= 10	a = a - 10
*=	a *= 10	a = a * 10
/=	a /= 10	a = a / 10
// =	a //= 10	a = a // 10
%=	a %= 10	a = a % 10
**=	a **= 10	a = a ** 10
&=	a &= 10	a = a & 10
=	a = 10	a = a 10
^=	a ^= 10	a = a ^ 10
>>=	a >>= 10	a = a >> 10
<<=	a <<= 10	a = a << 10

Identity operators

Operator	Description
is	Returns <i>True</i> if two variables point to the same object, <i>False</i> otherwise.
is not	Returns <i>True</i> if two variables point to different objects, <i>False</i> otherwise.

Membership Operators

Operator	Description
in	Returns <i>True</i> if the value/variable is found in the sequence, <i>False</i> otherwise.
not in	Returns <i>True</i> if the value/variable is not found in the sequence, <i>False</i> otherwise.

Truth Table

A	B	$\sim A$	$A \& B$	$A \mid B$	$A \wedge B$
		not A	A and B	A or B	[A xor B]
0	0	1	0	0	0
0	1	1	0	1	1
1	0	0	0	1	1
1	1	0	1	1	0

Precedence	Operator	Operator name
Highest	**	Exponential
	+x, -x, ~x	Sign: positive, negative Bitwise negation
	*, /, //, %	Multiplication, division, integer division, remainder
	+, -	Addition, subtraction
	<<, >>	Bitwise left shift, right shift
	&	Bitwise and
	^	Bitwise exclusive or (xor)
		Bitwise or
	==, !=, <, <=, >, >=, is, is not	Comparison, identity
	not	Logical negation
	and	Logical and
Lowest	or	Logical or

Operator precedence

Basic algorithm structures

- ▶ Sequence
 - set of instructions
- ▶ Selection
 - if-else statements
- ▶ Iteration
 - loops

Sequence

- ▶ The simplest control mechanism is actually a set of instructions executed in a predetermined order.
- ▶ Each instruction is written on a separate line.

instruction₁

instruction₂

...

instruction_n

Conditional statement (branching, selection)

- ▶ Examine a condition and execute different instructions depending on its truth value.

if condition:

statement1

[else:

statement2]

- ▶ The expression in the condition must always return a logical value, or something convertible to a boolean.
- ▶ If the condition is true, it executes *statement1*.
- ▶ Otherwise, if there is an *else* branch, it executes *statement2*.
- ▶ If there is no *else* branch and the condition is false, nothing happens.

Conditional statement

if condition:

statement1

statement2

else:

statement3

statement4

Exercise

- ▶ Write a program that determines whether the input number is even or odd.

Solution

```
x = int(input("x="))  
if x % 2 == 0:  
    print("Even!")  
else:  
    print("Odd!")
```

Exercise

- ▶ Input an integer and decide whether it ends with the digit 4 or not.

Solution

```
n = int(input("n = "))  
if n % 10 == 4:  
    print("Ends with 4.")  
else:  
    print("Does not end with 4.")
```

Exercise

- ▶ Write a program that determines whether a triangle can be constructed from three segments. If possible, calculate and display the area of the triangle.

Solution

```
import math

a = int(input("a = "))
b = int(input("b = "))
c = int(input("c = "))
if a + b > c and a + c > b and b + c > a:
    print("The triangle can be constructed!")
    p = (a + b + c)/2
    A = math.sqrt(p * (p - a) * (p - b) * (p - c))
    print("A = {0:.2f}".format(A))
else:
    print("The triangle cannot be constructed!")
```


Conditional statement

```
if condition1:  
    statement1  
elif condition2:  
    statement2  
elif condition3:  
    satement3  
...  
...  
else:  
    statementn
```

- ▶ the else branch may be omitted
- ▶ any number of elif branches can be used

Conditional statement

- ▶ The program executes the first branch whose condition is satisfied.
- ▶ If no condition is satisfied and there is an *else* branch, it executes the instructions there.
- ▶ If multiple logical expressions are combined with **and** / **or** operators:
 - **AND (and)** operator, if the first expression is false, then the second is ignored, the result is false anyway.
 - **OR (or)** operator, if the first expression is true, the second expression is ignored, the result is true anyway.

Exercise

- ▶ Write a program that reads two integers and prints:
 - if they are equal, or
 - if not, which one is greater.

Solution

```
a = int(input("a="))
b = int(input("b="))
if a == b:
    print("Equal!")
elif a > b:
    print("Greater: ", a)
else:
    print("Greater: ", b)
```

Exercise

- ▶ Ask the user "How many hours?".
- ▶ The program greets the user depending on the input:
 - Good Morning (5:00–11:59),
 - Good Afternoon (12:00–17:59),
 - Good Evening (18:00–21:59),
 - Good Night (22:00–4:59)

Solution

```
hour = int(input('hour = '))

if hour >= 5 and hour < 12:
    print("Good Morning!")
elif hour >= 12 and hour < 18:
    print("Good Afternoon!")
elif hour >= 18 and hour < 22:
    print("Good Evening!")
elif (hour >= 22 and hour < 24) or (hour >= 0 and
hour < 5):
    print("Good Night!")
else:
    print("Invalid value!")
```

The Pass statement

- ▶ Acts as a placeholder.
- ▶ Does nothing.
- ▶ Used as an empty statement.

```
n = int(input("n = "))
if n <= 100:
    pass
else:
    print("Invalid value!")
```

Exercise – Homework 1

- ▶ **Create a simple calculator:** Write a program that inputs two integers and an operator symbol (+, -, *, /, //, %).
- ▶ Perform the calculation according to the operation symbol.
- ▶ Show an error message if the operator is invalid or if division by zero occurs.
- ▶ **Input:** a, b, op
- ▶ **Output:** a op b = result
- ▶ **For example:**
 - $5 + 2 = 7$
 - $5 - 2 = 3$
 - $5 * 2 = 10$
 - $5 / 2 = 2.50$
 - $5 // 2 = 2$
 - $5 \% 2 = 1$

Exercise – Homework 2

- ▶ Write a program that reads the coefficients of a quadratic (second-degree) equation and prints its roots.
- ▶ **Input:** a, b, c
- ▶ **Output:** x_1, x_2

$$ax^2 + bx + c = 0$$

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

Exercise – Homework 3

- ▶ Write a program that prints the number of days in a given month of a given year.
- ▶ **Input:**
 - Year
 - Month
- ▶ **Output:**
 - Days in that month e.g.:
 - 2025, 9 -> 31
 - 2025, 4 -> 30
 - 2021, 2 -> 28
 - 2000, 2 -> 29